

Model Validation

Statistical Foundations of AI and ML, Module 5

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Concept of Model Validation

- ▶ We have discussed so many ways to fit a model on given data $X = \{x_1, \dots, x_N\}$
- ▶ Choose a model structure (eg. IID Gaussian, Gaussian Mixture Model, Hidden Markov Model), and then estimate their various parameters
- ▶ Or, fit a non-parametric or semi-parametric model using Parzen Window, Kernels or Copulas
- ▶ Given a model p and a set of parameters θ , we can calculate likelihood $p_\theta(X)$
- ▶ Given models $p_1, \dots, p_M$ with parameters $\theta_1, \dots, \theta_M$, we can compare them using $p_{1,\theta_1}(X), \dots, p_{M,\theta_M}(X)$

Hypothesis Behind Models

- ▶ Oftentimes we essentially make hypothesis about the data
 - ▶ The data are IID
 - ▶ The dataset follows a Gaussian Mixture Model
 - ▶ The dataset follows Bernoulli distribution with parameter 0.8
 - ▶ Two sets of data follow Gaussian distribution and have equal means
 - ▶ The data follows a relation $y = 2.1 * x_1 + 0.8 * x_2 - 5.1$
- ▶ But these are hypothesis we have proposed (maybe on some basis). How valid are they?
- ▶ **Hypothesis Testing**: carry out some statistical tests on the data to see if the hypothesis are valid!
- ▶ Basic idea: If hypothesis is wrong, then any statistic of the dataset must have low probability under it!

Motivating Example

- ▶ I have a coin. My default (NULL) hypothesis $\mathcal{H}_0$: it is unbiased, i.e. $p = 0.5$
- ▶ We carry out 10 coin tosses, results: 7H 3T
- ▶ Does this validate or invalidate my default hypothesis?
- ▶ If $\mathcal{H}_1$ was true, $p(X) \approx 0.11$ (quite low)
- ▶ It is unlikely, but not impossible!
- ▶ But if data were 9H 1T, $p(X) \approx 0.01$, which is pretty much impossible!
- ▶ In that case, we may have to rule out the original hypothesis!
- ▶ We can consider a cut-off probability α (level of significance), such that we can reject $\mathcal{H}_0$ if $p(X) < \alpha$ under $\mathcal{H}_0$
- ▶ But what can be an alternate hypothesis $\mathcal{H}_1$?

Criteria for Rejection

- ▶ We can reject $\mathcal{H}_0$ in favour of an $\mathcal{H}_1$, if the observations are more likely under it
- ▶ Alternate hypothesis can be vague, eg. *the coin is biased, i.e. $p \neq 0.5$*
- ▶ This is a *two-sided alternative hypothesis*, as it includes high bias ($p > 0.5$) or low bias ($p < 0.5$)
- ▶ Given observations 9H1T, we may reject $\mathcal{H}_0$ in favour of $\mathcal{H}_1$
- ▶ Alternatively, we can have a *one-sided alternative hypothesis* like $p < 0.5$
- ▶ In this case, the observations (9H 1T) are even more unlikely under $\mathcal{H}_1$ than $\mathcal{H}_0$!

Hypothesis Testing Framework

- ▶ Choose the null and alternate hypotheses $\mathcal{H}_0$ and $\mathcal{H}_1$
- ▶ Choose a test statistic $T(X)$ for the data X , whose statistical properties can be calculated under $\mathcal{H}_0$ (and perhaps $\mathcal{H}_1$)
- ▶ Choose the test criteria, i.e. a **rejection region** R , such that you can reject $\mathcal{H}_0$ if $T(X) \in R$
- ▶ Choose the **level of significance** α , which is the probability of the observations lying in the rejection region if $\mathcal{H}_0$ is true
- ▶ Note that $p_{\mathcal{H}_0}(T(X) \in R) = \alpha$
- ▶ **Type-1 error**: Reject $\mathcal{H}_0$ although it is true i.e. $T(X) \in R$ (α too high)
- ▶ **Type-2 error**: Accept $p_{\mathcal{H}_0}$ although it is false, i.e. $(T(X) \notin R)$ (α too low)
- ▶ R should have low probability under $\mathcal{H}_0$ and high(er) probability under $\mathcal{H}_1$

Examples

- ▶ Given a coin, $\mathcal{H}_0 : p = 0.5$ and $\mathcal{H}_1 : p > 0.5$ (one-sided)
- ▶ Let $T_N(X)$ denote the fraction of heads in N tosses
- ▶ If $\mathcal{H}_0$ holds, N tosses should produce roughly equal number of heads and tails, but under $\mathcal{H}_1$ there can be significantly more heads than tails
- ▶ Rejection Region: $T_N(X) - 0.5 \geq \epsilon$ where we need to choose ϵ
- ▶ If ϵ is too low (close to 0), then even moderate results like 6H4T may not be acceptable (risk of type 1 error)
- ▶ If ϵ is too high (close to 0.5), then even extreme results like 9H1T are acceptable (risk of type 2 error)
- ▶ Under $\mathcal{H}_0$, as $p_{\mathcal{H}_0}(T_{10}(X) - 0.5 \geq 0.4) = 0.011$
- ▶ So if $R = \{T_{10}(X) = 0.9\}$, we can accept $\mathcal{H}_0$ at 1% level of significance (but not at, say 5% level of significance)

Examples

- ▶ Alternatively we specify R in terms of α (instead of ϵ), i.e. we will reject $\mathcal{H}_0$ only if the probability of the data is less than α under $\mathcal{H}_0$
- ▶ In that case, R has to be understood depending on $\mathcal{H}_1$
- ▶ If $\mathcal{H}_1 : p \neq 0.5$ (two-sided), then we must consider very high as well as very low values of $T_N(X)$ for rejection
- ▶ So $R = \{R_1, R_2\}$ such that $R_1 : \{T_N(X) \leq 0.5 - \epsilon_1\}$, and $R_2 : \{T_N(X) \geq 0.5 + \epsilon_2\}$ with $p_{\mathcal{H}_0}(R_1) + p_{\mathcal{H}_0}(R_2) = \alpha$
- ▶ Usually we can consider: $p_{\mathcal{H}_0}(R_1) = p_{\mathcal{H}_0}(R_2) = \frac{\alpha}{2}$
- ▶ **Exercise:** If $N = 10$, find R_1 and R_2 at 5% and 1% levels of significance

Examples

- ▶ Suppose $x_1 \dots x_N \sim \mathcal{N}(\mu, \sigma^2)$ where σ is known, and $\mathcal{H}_0 : \mu = \mu_0$
- ▶ One-sided alternate hypothesis: $\mathcal{H}_1 : \mu > \mu_0$
- ▶ Test statistic: $T_N(X) = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{N}}$
- ▶ Under $\mathcal{H}_0$, $T_N(X) \sim \mathcal{N}(0, 1)$
- ▶ $R : T_N(X) > c$ (i.e. $\bar{X}$ is far higher than μ_0)
- ▶ Clearly, $\alpha = p_{\mathcal{H}_0}(T_N(X) > c) = 1 - \Phi(c)$, i.e. $c = \Phi^{-1}(1 - \alpha)!$
- ▶ Two-sided alternate hypothesis: $\mathcal{H}_1 : \mu \neq \mu_0$
- ▶ In that case, $R : |T_N(X)| > c$, i.e. $R_1 : T_N(X) < -c$, $R_2 : T_N(X) > c$
- ▶ By symmetry properties of Gaussian distribution, $\alpha = 2\Phi(-c)$, i.e. $c = -\Phi^{-1}(\alpha/2)$

- ▶ Clearly, we tend to reject the null hypothesis if the observations are **extreme**, i.e. unlikely under the null hypothesis
- ▶ **p-value** is the probability of obtaining results at least as extreme as the observations, under the assumption that the null hypothesis is correct
- ▶ Given $X = 8H2T$ from a coin under $\mathcal{H}_0 : p = 0.5$, p-value is the probability of $\{8H2T, 9H1T, 10H0T\}$
- ▶ The p-value represents the smallest level of significance α at which we will reject $\mathcal{H}_0$
- ▶ Suppose $X = \{x_1, \dots, x_N\} \sim \mathcal{N}(\mu, \sigma_0^2)$ where $\mathcal{H}_0 : \mu = \mu_0$ and $\mathcal{H}_1 : \mu \neq \mu_0$
- ▶ Clearly, $T_N = \left| \frac{\bar{X} - \mu_0}{\sigma_0^2 / \sqrt{N}} \right|$ under $\mathcal{H}_0$, whose observed value is t_N
- ▶ p-value is then $2 * \min(p_{\mathcal{H}_0}(T_N > t_N), p_{\mathcal{H}_0}(T_N < t_N))$

Likelihood Ratio Tests

- ▶ Suppose we did not have the null hypothesis. How likely would the observations have been?
- ▶ Eg. in the coin-toss case, what is the probability of $6H4T$ irrespective of p (i.e. marginalizing p)?
- ▶ Define $\lambda(X) = \frac{p_{\mathcal{H}_0}(X)}{p(X)}$, , reject $\mathcal{H}_0$ if $\lambda(X) < c$
- ▶ Logic: the $\mathcal{H}_0$ should be able to concentrate a significant share of the overall probability of the data!
- ▶ So far, we have considered probability of observations under $\mathcal{H}_0$ only, but how about $\mathcal{H}_1$?
- ▶ Define $\lambda(X) = \frac{p_{\mathcal{H}_0}(X)}{p_{\mathcal{H}_1}(X)}$
- ▶ Rejection criteria: if $\lambda(X) < c$, where c is some value below 1 (i.e. data is significantly more likely under alternate hypothesis)

Chi-square Test

- ▶ How well does your model f fit the data?
- ▶ This test allows us to compare the data histogram $h(X)$ with the fitted model $f(x)$
- ▶ h divides the observations into K intervals $(y_1, \dots, y_K)$
- ▶ Define test statistic $T_K(f) = N * (\sum_{k=1}^K \frac{(h(k)-f(k))^2}{f(k)})$ where $h(k)$ is the number of observations in the k -th interval and $f(k)$ is its expected value according to f
- ▶ i.e. $h(k) = \frac{\sum_{i=1}^N I(x_i \in y_k)}{N}$ and $f(k) = \int_{y_k} f(x) dx$
- ▶ Theorem: $T_K(f) \sim \chi^2(K - m)$ where m is the number of parameters in f
- ▶ Given a level of significance α , we can identify the critical value χ_0 , such that $\chi_{K-m}^2(T_K(f) > \chi_0) = \alpha$
- ▶ We can reject the null hypothesis if the calculated value of $T_K(f)$ exceeds χ_0

Chi-square Test

- ▶ We can also use χ^2 -test for checking the homogeneity of a population, or for independence between two variables
- ▶ Consider two attributes X_1, X_2 (eg. height and weight) of N individuals.
- ▶ They are discretized into L and M values, and we have a $M \times N$ table Y with $Y_{l,m} = \sum_{i=1}^N I(X_{1i} \in A_{1l}, X_{2i} \in A_{2m})$
- ▶ $\mathcal{H}_0$: the attributes are independent (or one attribute is similarly distributed for all values of the other attribute)
- ▶ Under $\mathcal{H}_0$, $E(Y_{lm}) = N * p_{1l} * p_{2m}$ where $p_{1l} = \frac{\sum_{i=1}^N I(X_{1i} \in A_{1l})}{N}$ and $p_{2m} = \frac{\sum_{i=1}^N I(X_{2i} \in A_{2m})}{N}$ i.e. marginals
- ▶ Test statistic
$$T_{X_1, X_2} = \sum_{l=1, m=1}^{L, M} \frac{(Y_{lm} - E(Y_{lm}))^2}{E(Y_{lm})} \sim \chi^2((m-1) * (l-1))$$

Permutation Test

- ▶ $\{X_1, \dots, X_M\} \sim F, \{Y_1, \dots, Y_N\} \sim G$
- ▶ $\mathcal{H}_0 : F = G, \mathcal{H}_1 : F \neq G$
- ▶ Idea: Under $\mathcal{H}_0$, their sample means $\bar{X}, \bar{Y}$ should be close,
- ▶ Further, if we consider a random partition of $\{X, Y\}$, their sample means should be close to $\bar{X}, \bar{Y}$!
- ▶ Create joint dataset $Z = \{X_1, \dots, X_M, Y_1, \dots, Y_N\}$ and $L = \{1, \dots, (M \text{ times}), 1, 2, \dots, (N \text{ times}), \dots, 2\}$
- ▶ Test statistic $g(L, Z) = \left| \frac{\sum_i Z_i I(L_i=1)}{M} - \frac{\sum_i Z_i I(L_i=2)}{N} \right|$
- ▶ Define $p = \frac{1}{Q} \sum_{\pi} I(g(L_{\pi}, Z) > g(L, Z))$ where L_{π} denotes various permutations of L *where Q is no. of permutations*
- ▶ In how many permutations of L is the difference of means greater than the observed difference between the means??
- ▶ Under $\mathcal{H}_0, p \sim U(0, 1)$, so reject $\mathcal{H}_0$ at LOS α if $p > \alpha$

Information Criteria for Model Selection

- ▶ A model tries to fit the data, i.e. $p_{MODEL}(X)$ should be high
- ▶ But if the model has too many parameters, it may overfit the dataset!
- ▶ Necessary to trade-off likelihood and parameter size
- ▶ Bayesian Information Criteria (BIC):
 $2\log(P(X|M, \hat{\theta}_M)) - d_M \log N$ where $\hat{\theta}_M$ is the ML-estimate of model M 's parameters, and d_M is the number of parameters
- ▶ Akaike Information Criteria (AIC): $2\log(P(X|M, \hat{\theta}_M)) - 2d_M$
- ▶ Given a set of models, these criteria can be used to choose the optimal one
- ▶ McFadden's $R^2 = 1 - \frac{\log(L_M)}{\log(L_0)}$ where L_0 is the data probability under reference model (eg. model that actually generated the data), L_M is data probability under candidate model M